

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 4. Operators and Expressions

Lecture: 29. Program using Expressions

Section 1: Lecture Summary

This lecture introduced how to write a simple Python program using expressions, specifically to calculate the area of a rectangle. The instructor began by explaining the mathematical concept of the area of a rectangle, defining length and breadth, and the formula for area ($\text{length} \times \text{breadth}$). Then, a complete Python program was demonstrated where variables for length and breadth were assigned values, the area was calculated using multiplication, and the result was printed. The program was executed inside a Jupyter Notebook. Emphasis was placed on writing complete programs, practicing by typing and executing code, and preparing for more complex programming challenges in future lectures.

Section 2: Key Concepts and Explanations

- Expressions in Programming

- An expression combines variables, operators, and values to perform a computation.
- Example: `area = length * breadth` is an expression calculating area.

- Variables and Assignment

- Variables store data values.
- Assignment operator `=` assigns a value to a variable.
- Example: `length = 15` assigns the integer 15 to variable `length`.

- Arithmetic Operators

- Used to perform mathematical operations.

- Multiplication (`*`) is used here to calculate the rectangle's area.

- Python Program Structure

- A program consists of statements executed sequentially.
- Typical steps: define variables, perform calculations, output results.

- Output with `print()`

- Displays information on the screen.
- Multiple arguments can be passed separated by commas, e.g., printing label and value.

- Using Jupyter Notebook

- An interactive environment to write, run, and test Python code.
- Run code cells with Shift+Enter.

Section 3: Example Code and Use Cases

Assign length and breadth of the rectangle

Calculate the area using multiplication

Print the result with a descriptive label

```
length = 15
breadth = 5

area = length * breadth

print('Area:', area)
```

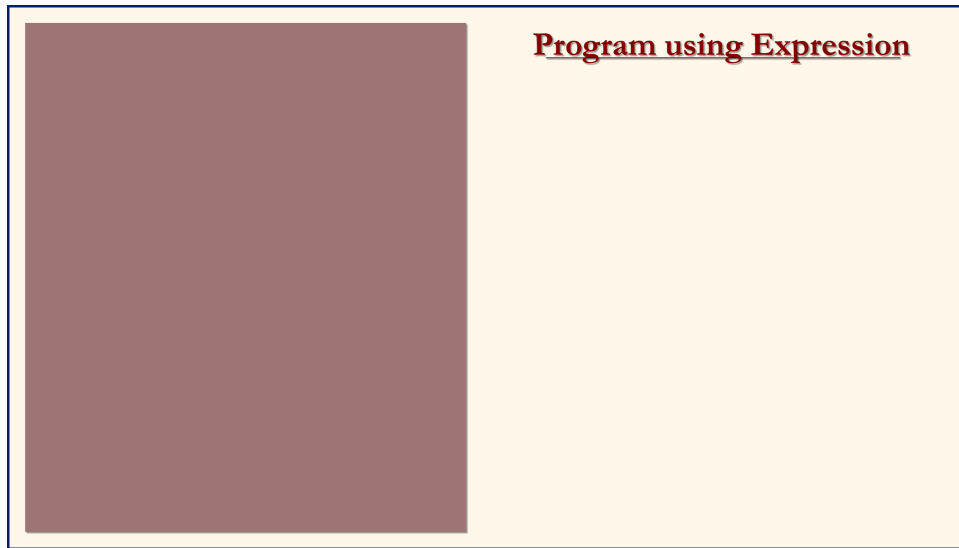
Explanation: This simple program assigns values 15 and 5 to `length` and `breadth`, computes `area` by multiplying them, and prints the result as "Area: 75".

Section 4: Key Takeaways

- Expressions use variables and operators to compute values.
- Variables are defined using assignment (`=`).
- Calculations can be done using arithmetic operators like `+`.
- `print()` displays output; it can show text and variables together.
- Writing complete programs involves defining inputs, processing, and output steps.
- Jupyter Notebook is a useful tool for writing and executing Python code interactively.
- Practice writing and running programs to build programming skills progressively.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Program using Expression

How to write a Program?

Program using Expression

How to write a Program?

Program to Find Area of Rectangle



Program using Expression
How to write a Program?

Program to Find Area of Rectangle



Program using Expression
How to write a Program?

Program to Find Area of Rectangle

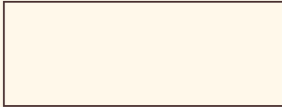
length



Program using Expression
How to write a Program?

Program to Find Area of Rectangle

breadth



length



Program using Expression
How to write a Program?

Program to Find Area of Rectangle

breadth

15



length



Program using Expression
How to write a Program?

Program to Find Area of Rectangle

breadth

5

15

length

area



Program using Expression
How to write a Program?

Program to Find Area of Rectangle

breadth

5

15

length

area = length * breadth

Python Program

Program using Expression

How to write a Program?

Program to Find Area of Rectangle

breadth 5

15

length

area = length * breadth

Python Program

```
length = 15
```

Program using Expression

How to write a Program?

Program to Find Area of Rectangle

breadth 5

15

length

area = length * breadth

Python Program

```
length = 15  
breadth = 5
```

Program using Expression

How to write a Program?

Program to Find Area of Rectangle

breadth 5

15

length

```
area = length * breadth
```

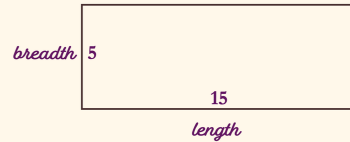
Python Program

```
length = 15  
breadth = 5  
area = length * breadth
```

Program using Expression

How to write a Program?

Program to Find Area of Rectangle



```
area = length * breadth
```

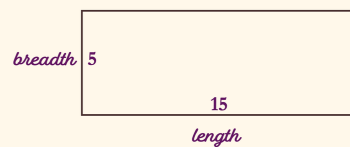
Python Program

```
length = 15  
breadth = 5  
area = length * breadth  
print('Area:' , area)
```

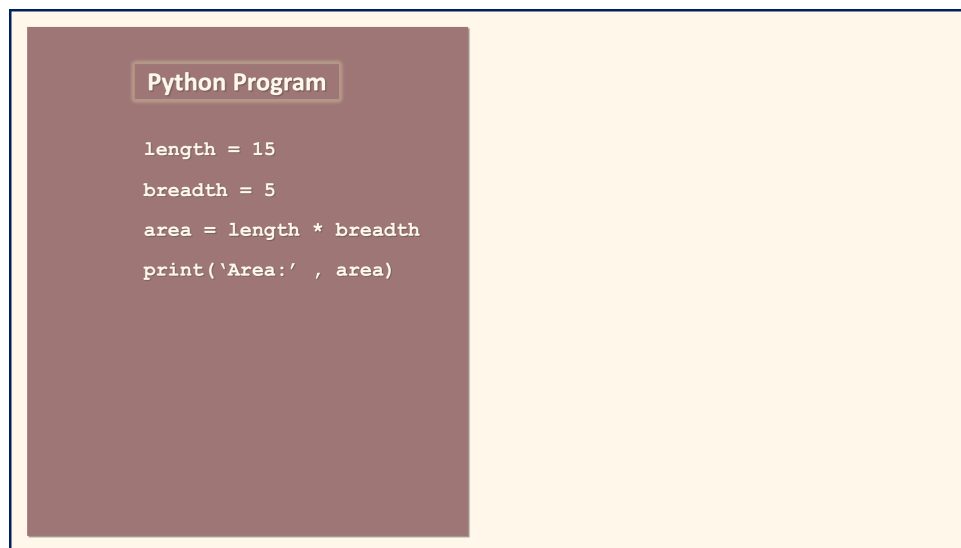
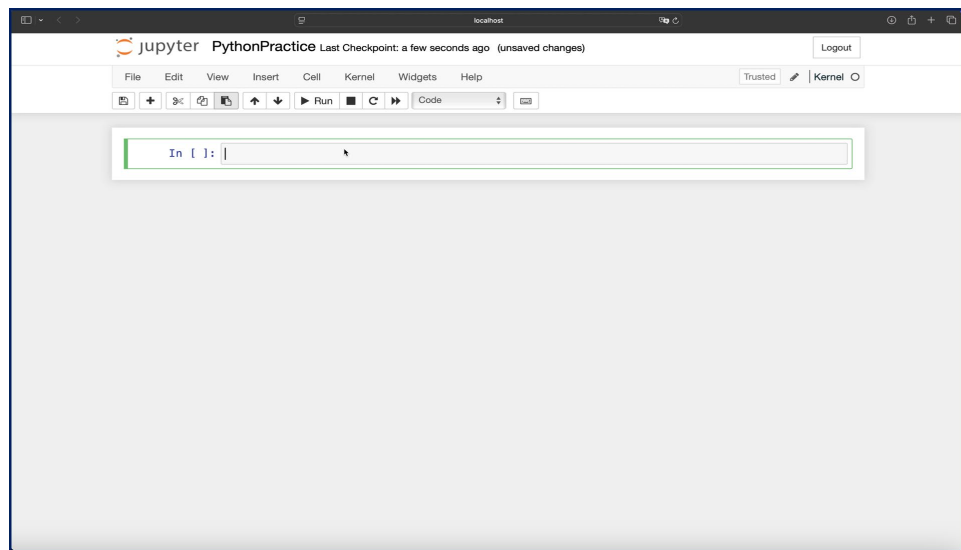
Program using Expression

How to write a Program?

Program to Find Area of Rectangle

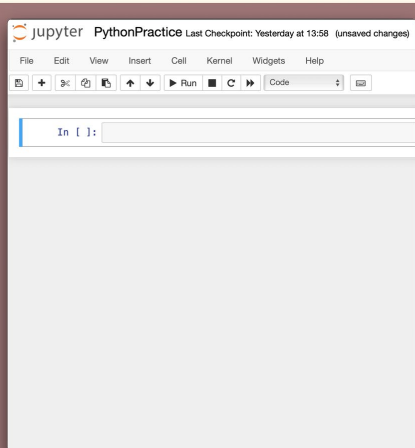


```
area = length * breadth
```



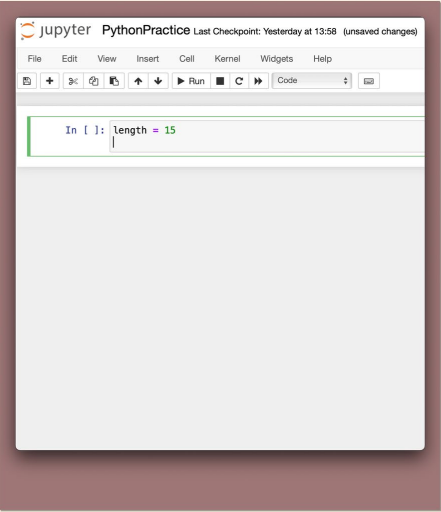
Python Program

```
length = 15  
breadth = 5  
area = length * breadth  
print('Area:' , area)
```



Python Program

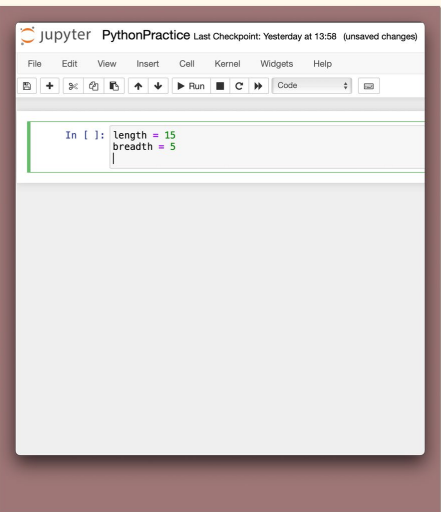
```
length = 15  
breadth = 5  
area = length * breadth  
print('Area:' , area)
```



The Jupyter Notebook interface shows a code cell with the text `In [ ]: length = 15`. The notebook title is "PythonPractice" and it indicates "Last Checkpoint: Yesterday at 13:58 (unsaved changes)".

Python Program

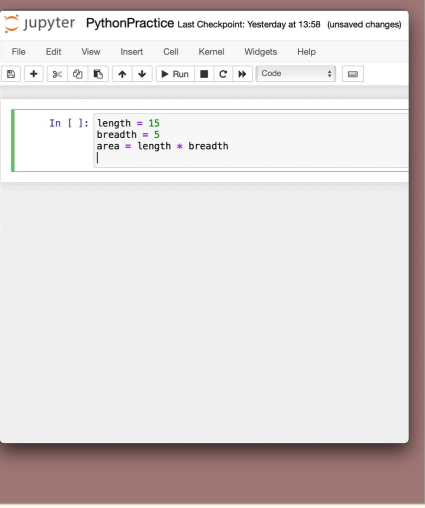
```
length = 15  
breadth = 5  
area = length * breadth  
print('Area:' , area)
```



The Jupyter Notebook interface shows a code cell with the text `In []: length = 15
breadth = 5`. The notebook title is "PythonPractice" and it indicates "Last Checkpoint: Yesterday at 13:58 (unsaved changes)".

Python Program

```
length = 15  
breadth = 5  
area = length * breadth  
print('Area:' , area)
```

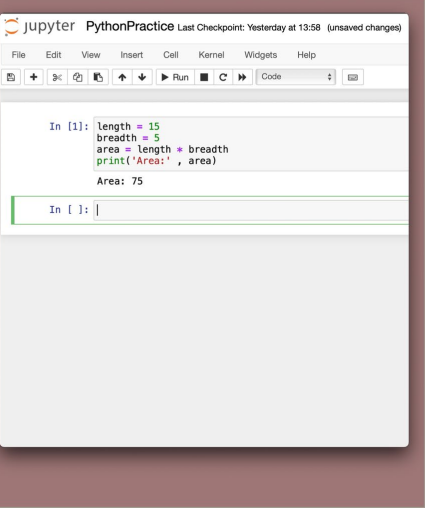


The Jupyter Notebook interface displays a single code cell with the following Python code:

```
In [ ]: length = 15  
        breadth = 5  
        area = length * breadth  
        |
```

Python Program

```
length = 15  
breadth = 5  
area = length * breadth  
print('Area:' , area)
```



The Jupyter Notebook interface shows the code cell from the previous image, now with its output displayed below the code:

```
In [1]: length = 15  
        breadth = 5  
        area = length * breadth  
        print('Area:' , area)  
        Area: 75  
In [ ]: |
```

Python Program

```
length = 15  
breadth = 5  
area = length * breadth  
print('Area:' , area)
```

